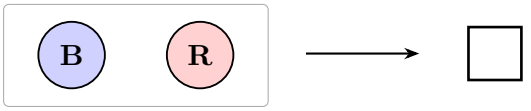
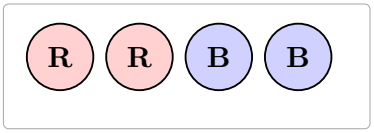
Use the same bag of three red counters and one blue. Draw twice in order, independently, giving every counter the same chance each time. Return and mix after each draw.



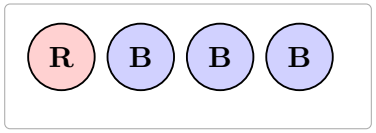
A sample rule: BR gives a square.

A rule gives each color pair a square, a circle, or no shape. A fair rule must sometimes give each shape, and give them equal chances when a shape comes out.

Problem 3: Keep your rule for each bag below. Which bags give the two shapes the same chance? Show a reason that does not depend on a lucky run.

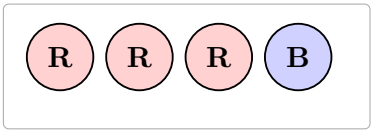


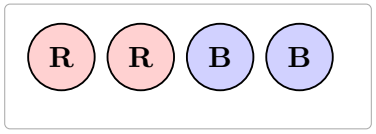
two red, two blue



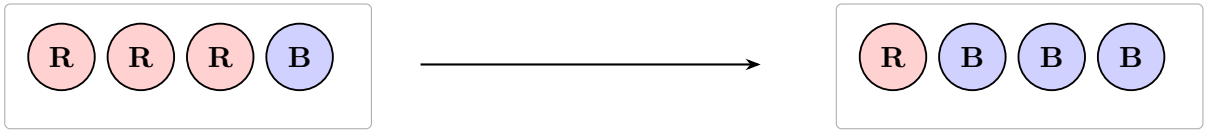
one red, three blue

Problem 4: You want shapes to come out often. Would you choose the three-red bag or the two-red bag? Give an exact reason.





Problem 5: Change bags between draws: use the left bag first and the right bag second. Does your rule still give the shapes the same chance?



first draw second draw

Problem 6: Can your rule promise at least one shape within six pairs? Make a six-pair story that settles your answer.

Problem 7: Can any fair rule give a shape on every pair from the original three-red, one-blue bag? Explain your answer.